

DevDash'26

Problem Statement Universal College Lanka

1. Introduction

Universal College Lanka (UCL) is inviting development teams to design a solution for a growing gap in how students access campus information; this document sets out the business needs behind that initiative.

2. Business Background

Students currently rely on a mix of disconnected channels like WhatsApp groups, lecturers, notice boards, societies and word of mouth to stay informed about university life, with no single channel treated as authoritative. This fragmentation spans announcements, events, lost property, campus services and society activity, adding to the workload of administrative staff who often become the fallback source.

This is a structural gap rather than an incidental one, there has never been a shared channel built for this purpose, so information has simply spread across whichever tools were convenient at the time

3. Problem Statement

Universal College Lanka requires a digital solution that gives students one reliable, trusted channel for campus-related information and services, in place of the disconnected mix they rely on today.

UCL is asking participating development teams to decide which parts of this problem to address, and to design and prototype a solution accordingly. No specific form of solution is prescribed, teams are free to propose any approach, provided it responds to the business needs described in this document.

4. Project Objectives

Beyond solving the immediate problem, UCL sees this initiative as contributing to a small number of broader outcomes:

- A more informed and engaged student body.
- Greater visibility for events, societies and campus services.
- Less time spent by staff and students searching for basic information.
- A stronger long-term foundation the university can build on beyond the hackathon.

These outcomes are intentionally broad. Teams are free to decide which aspects of the problem contribute most to them, and to prioritise accordingly.

Marks next to each requirement (Sections 5 and 6) reflect relative importance out of a combined 100, scaled to the 45 marks for Requirement Coverage in [Section 10](#). See [Section 10](#) for how solutions are assessed at each classification level.

5. Business Requirements

The following business requirements describe what the university expects a solution to accomplish. They do not prescribe how a solution should be built, and are not listed in order of priority.

BR1 – Unified Access: A single way for students to access campus-related information that is currently spread across disconnected channels. **(5 Marks)**

BR2 – Targeted Announcements: Timely delivery of official university announcements and other content relevant to a student's specific faculty, year group or programme, distinct from general university-wide information. **(3 Marks)**

BR3 – Event Visibility: Visibility of campus events, covering both university-organised events and events publicised by student organisers. **(3 Marks)**

BR4 – Event Interest: A way for students to register interest in attending an event, so organisers can gauge expected turnout. **(1 Marks)**

BR5 – Society Visibility: A way for student societies to share information about their activities with the wider student body. **(3 Marks)**

BR6 – Society Sign-up: A way for students to sign up for or express interest in joining a student society. **(2 Marks)**

BR7 – Lost & Found: A way for students and staff to report and search for lost and found items. **(3 Marks)**

BR8 – Classroom Booking: A way for students to check classroom availability and request a room for study or group work, without going through administrative staff directly. **(5 Marks)**

BR9 – Academic Support: A way for students to request academic support, such as study groups, peer tutoring, or mentorship. **(3 Marks)**

BR10 – FAQ Access: Access to answers for commonly asked questions, reducing the volume of repetitive enquiries directed to administrative staff. **(2 Marks)**

BR11 – Content Maintenance: A way for staff, society representatives, or other authorised contributors to publish and keep accurate the information they're responsible for. **(4 Marks)**

BR12 – Access Levels: Different levels of access for students and staff — students can view information, while different staff roles (such as academic, society, finance, or administrative) can add, update, or manage it. **(6 Marks)**

BR13 – Academic Calendar: Visibility of academic dates and deadlines relevant to students, such as exam periods, add/drop dates, or semester milestones. **(3 Marks)**

BR14 – Student Onboarding: Access to information that supports new or first-year students in transitioning into university life. (2 Marks)

BR15 – Emergency Communication: A way for the university to share safety-related or emergency information with students promptly. (3 Marks)

BR16 – Schedule Changes: A way for the university to communicate closures or schedule changes due to unforeseen circumstances, such as weather or public holidays. (1 Mark)

BR17 – Feedback Loop: A way for students to give feedback or raise questions not already addressed, supporting continual improvement of the information provided. (1 Mark)

BR18 – Volunteering Opportunities: Visibility of volunteering or community engagement opportunities available to students. (1 Mark)

BR19 – Alumni Engagement: A way for alumni-related activities or opportunities to be communicated to current students, where relevant. (1 Mark)

BR20 – Job & Internship Visibility: Visibility of part-time job, internship or placement opportunities relevant to students. (3 Marks)

BR21 – Facility Issue Reporting: A way for students to report maintenance issues or facility problems they encounter on campus. (2 Marks)

BR22 – Staff Directory: Access to a directory of staff or departmental contacts relevant to student needs. (2 Marks)

BR23 – Financial Support Info: Access to information about scholarships, financial aid, or fee-related support available to students. (3 Marks)

BR24 – Sports & Recreation: Access to sports and recreational facility information, including how to access or book them. (2 Marks)

BR25 – Dining Information: Access to canteen or dining menu and operating hours information. (1 Mark)

BR26 – Printing Services: Access to printing, copying, or stationery service information available on campus. (1 Mark)

BR27 – Textbook Exchange: A way for students to exchange or pass on second-hand textbooks or study materials. (1 Mark)

BR28 – Guest Lectures: A way for the university to publicise guest lectures, industry talks, or workshops, distinct from regular campus events. (1 Mark)

BR29 – Wellbeing Support: Access to information about student wellbeing and counselling support services. (3 Marks)

BR30 – IT Support Info: Access to information about IT or technical support available to students. (2 Marks)

BR31 – Library Resources: Access to library resources, opening hours, or borrowing services.
(2 Marks)

BR32 – Student Life Highlights: A way for the university to highlight past events or achievements, giving students visibility into ongoing student life and culture.
(1 Marks)

BR33 – AI Assistant: A way for students to get help finding information or answers to their questions or guide them through the solution quickly for ease of access through an assistant that can respond to natural, conversational queries using AI, rather than requiring them to browse or search manually.
(9 Marks)

6. Non-Functional Requirements

The following non-functional requirements describe the qualities the solution is expected to exhibit. Development teams should interpret these in a manner appropriate to their chosen solution approach.

NFR1. Usability: Easy enough for a first-time student user - including those with varying levels of digital literacy - to understand and use without prior training or guidance.
(2 Marks)

NFR2 – Performance & Scalability: Responsive enough that user actions feel quick and comfortable, and able to remain usable as the number of users or volume of content grows, including during peak periods such as the start of a semester.
(3 Marks)

NFR3 – Reliability & Availability: Reliable and available enough that students can depend on it for time-sensitive information, at times that suit typical university activity patterns.
(1 Mark)

NFR4 – Security & Privacy: Protective of the personal information of students and staff, restricting sensitive functions — such as publishing official announcements — to authorised users only.
(2 Marks)

NFR5 – Maintainability: Maintainable by future development or administrative teams, so the university can sustain and update it beyond the hackathon.
(4 Marks)

NFR6 – Robustness: Reasonably tolerant of incorrect or incomplete information entered by users, without failing outright.
(3 Marks)

7. Assumptions

Development teams may make the following reasonable assumptions where information has not been explicitly provided:

- Students and staff have reasonable access to a smartphone, tablet or computer during their time on campus.
- The university will be responsible for populating and maintaining accurate information within any adopted solution.

- Not all categories of information described in this document need to be addressed equally; teams may choose to focus on a subset.
- Existing university systems, such as student records or timetabling systems, are not assumed to be available for integration during the hackathon.
- The university community includes students and staff with a range of technical backgrounds and comfort levels with digital tools.
- Any data used for demonstration purposes may be reasonably invented by the development team.

8. Constraints

The following business constraints apply to this initiative:

- The solution must be appropriate for use within a university environment and by a student and staff audience.
- The university does not currently have a dedicated technical team to maintain highly complex or specialised infrastructure.
- Any proposed solution should be realistic to introduce and sustain within the operational capacity of a university administrative office.
- The university has not allocated a specific technology stack or vendor; teams are free to determine an appropriate approach.
- Information published through the solution must remain appropriate for an academic institution.
- The university expects any solution to be respectful of student privacy and to handle personal information responsibly.

9. Expected Deliverables

At the conclusion of the project, UCL expects development teams to provide:

- A **working prototype**, developed under the provided GitHub organisation, that demonstrated the team's proposed solution to the business needs described in this document. The final commit made within the allocated time will be taken as the team's submission.
- A **report** documenting the team's approach to the problem, including their design process, testing outcomes, key decisions, and evidence of how the solution was developed and validated.
- A **product pitch** presenting the solution and its key functionality (slides are allowed). Slides can be prepared after the submission deadline

10. Judging Criteria

Phase 01: Code Review - 60%

Classification	Requirement Coverage & Depth (45)	Code Quality & Best Practices (9)	Report Quality (6)
Allocated Marks			
Excellent	Nearly all requirements work, with high complexity and strong non-functional quality (41–45)	Clean, consistent, well-commented code throughout; no magic numbers or clutter (8-9)	Report clearly documents design, architecture and testing with strong evidence throughout (5–6)
Good	Most requirements work, good complexity but some simpler features; minor non-functional gaps (31–40)	Reasonably organised code with occasional inconsistency or minor clutter (6–7)	Report covers most key areas: design, architecture, testing, with reasonable clarity (4)
Adequate	Roughly half work, mostly basic complexity; noticeable non-functional gaps (22–30)	Workable but inconsistent code; noticeable clutter, unclear naming, or missing comments (4–5)	Report is present but thin, covering only some required areas (2–3)
Fail	Few or none work; little feature depth or non-functional quality (0–21)	Code is disorganised, unreadable, or largely non-functional (0–3)	Report missing significant content or not usable (0–1)

Phase 02: Product Pitch - 40%

Classification	Requirement Completion, Understanding & Prioritisation (12)	Functionality & Live Demo (8)	Solution Defence (8)	Innovation & Creativity (4)	Presentation Delivery (8)
Allocated Marks					
Excellent	Clearly explains what was built, what was left out, and why, with strong justification of priorities (10–12)	Demo runs smoothly and clearly shows the claimed functionality working (7–8)	Confident and well-articulated defence; questions answered thoroughly with clear reasoning and strong technical understanding (4)	Genuinely distinctive or clever approach to the problem (7–8)	Clear, confident, well-structured delivery within time (7–8)
Good	Explains what was built and mostly why, with reasonable justification (7–9)	Demo mostly works, with only minor hiccups (5–6)	Most questions handled competently; minor gaps in explanation or technical depth (3)	Most questions handled competently; minor gaps in explanation or technical depth (3)	Generally clear and organised, with minor pacing issues (5–6)
Adequate	Describes what was built but with limited reasoning or prioritisation (5–6)	Demo has noticeable issues or only partially shows claimed functionality (3–4)	Some questions answered but responses lack clarity, depth or confidence in places (2)	Mostly conventional approach with little differentiation (3–4)	Delivery is unclear or disorganised; timing off (3–4)
Fail	Little understanding of the problem or reasoning shown (0–4)	Demo fails or doesn't reflect what was claimed (0–2)	Unable to defend the solution; responses incomplete or off-topic (0–1)	Generic or low-effort approach (0–2)	Difficult to follow; team struggles to communicate (0–2)

